A benchmark-validated language-model judge for measuring scientific corroboration of legislators' claims

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Abstract

Measuring whether the scientific record corroborates the empirical claims that legislators make has been limited by a costly human-coding bottleneck: deciding whether a study supports, refutes, or is silent on a claim has required expert annotators. We ask how far this one judgment can be automated and, crucially, how its trustworthiness can be established *without* new in-domain human labels. Our answer is benchmark-anchored validation: we score candidate stance judges against existing human-labeled scientific-claim-verification benchmarks (SciFact and Climate-FEVER) for the identical task. A proprietary frontier model (Claude Opus 4.8), used as an in-context judge, agrees with the expert gold on 80% of SciFact and 78% of Climate-FEVER pairs—comparable to reported frontier zero-shot results and approaching the human inter-annotator ceiling ($\kappa = 0.75$ on SciFact)—whereas among the open models that run locally only the natural-language-inference surrogate matches the judge, and only on clean SciFact pairs (0.74–0.76 vs. 0.80, overlapping intervals); the small instruction-tuned panel trails there, and all open models fall well short on the harder, more policy-like Climate-FEVER claims (0.49–0.67 vs. 0.78). Because the benchmark text is human-written, this accuracy is not an artifact of self-preference on the evidence side. We then propagate a small sample of expensive proprietary-judge labels through the cheap open surrogate over all pairs via prediction-powered inference, and apply the result to a decade of California Assembly press releases on artificial intelligence: Democrats engage with AI far more than Republicans, and the rate at which each party's claims are corroborated by the retrieved literature is similar across parties. We are deliberately explicit about scope. Only the *judging* step is validated, and only on benchmark pairings in which claim and evidence are already topically matched; the gold judge is proprietary, not local or free, so it is the open surrogate—not the whole pipeline—that runs without paid services; the headline corroboration rates depend on a proprietary generative retrieval step, whereas reproducible open keyword retrieval surfaces little claim-specific evidence; and an in-domain human audit—precluded by the no-human-coder design—remains the decisive missing test. The contribution is a transferable validation methodology, and an honest map of which components of autonomous evidence-corroboration measurement are open, reproducible, and trustworthy today, and which are not yet.

1 Introduction

Whether policymakers use scientific evidence, and whether the evidence they invoke actually supports the claims they make, is a central question for the study of evidence-based policymaking

(Oliver et al., 2022; Bogenschneider et al., 2019; Weiss, 1979). Answering it at scale has been blocked by a measurement bottleneck. Establishing that a given scientific study *supports*, *refutes*, or is *silent* on a specific empirical claim in a press release, floor speech, or bill is a judgment that has historically required trained human coders reading both the claim and the underlying science (Grimmer and Stewart, 2013). Expert coding is slow, expensive, and hard to reproduce, so studies of evidence use have been confined to small samples or coarse proxies, and cannot keep pace with the volume of modern political communication.

Large language models (LLMs) appear to dissolve this bottleneck. Across political science and computational social science, instruction-tuned LLMs now match or exceed trained human coders on stance detection, framing, and other annotation tasks, at a fraction of the cost (Gilardi et al., 2023; Ziems et al., 2024; Heseltine and Clemm von Hohenberg, 2024; Törnberg, 2025). This raises the possibility of an *autonomous* measurement pipeline—one that ingests raw political text, extracts the empirical claims, retrieves topically-relevant science, and judges whether that science supports each claim. The aspiration is to do this with no human annotators and, ideally, no proprietary services; how close one can actually get with open models is an empirical question we take up directly, and the answer turns out to be partial.

The obstacle is not building such a pipeline but *trusting* it. If the model that judges claim–evidence support is never checked against a human standard, its outputs are unfalsifiable: a fluent label is not necessarily a correct one (Bender et al., 2021), and the statistical machinery for turning noisy machine labels into valid population estimates—design-based supervised learning and prediction-powered inference (Egami et al., 2023; Angelopoulos et al., 2023), and the alternative-annotator test that licenses replacing a human coder (Calderon et al., 2025)—is all defined relative to trusted human ground truth. A second, subtler threat is circularity: when the claims, the retrieved references, and the judge are *all* produced by language models, an LLM judge may rate model-surfaced evidence as supportive because LLM evaluators are known to prefer model-generated and low-perplexity text—a documented *self-preference* bias (Panickssery et al., 2024; Zheng et al., 2023). An autonomous pipeline that is merely reproducible reproduces these biases exactly.

Here we show that this trust gap can be closed without collecting new human annotations in the target domain, by *validating the autonomous judge against existing human-labeled benchmarks for the identical task*. Scientific claim verification—deciding whether an abstract supports, refutes, or is neutral toward a claim—is precisely the support-assessment step of our pipeline, and it has mature, expert-annotated benchmarks: SciFact (Wadden et al., 2020) for expert scientific claims and Climate-FEVER (Diggelmann et al., 2020) for contested, real-world claims phrased in lay language. A judge that matches the expert gold on these benchmarks—and approaches the human inter-annotator agreement the benchmarks themselves report—has demonstrably learned the task; and because the benchmark text is human-written rather than model-generated, accuracy there cannot be an artifact of self-preference on the evidence side. Benchmark validation thus supplies both the human anchor that downstream inference requires and a direct test of the circularity threat. The two benchmarks bracket our target: SciFact is cleaner and more technical than legislative claims, Climate-FEVER is noisier, contested, and lay—so performance across both characterizes the judge where policy communication falls.

Using this strategy we study a measurement pipeline—claim extraction, reference retrieval from the open OpenAlex index (Priem et al., 2022), and stance judging—applied to a decade of legislative communication on artificial intelligence (AI). We make four contributions, and we scope each carefully. (1) A transferable *validation methodology*: benchmark-anchored evaluation that tests whether an automated stance judge can be trusted with no new in-domain human labels, by scoring it on existing human-labeled claim-verification benchmarks for the identical task. (2) A measurement finding with a sharp open/closed boundary: among open models that run locally, only the zero-shot NLI surrogate *matches* a proprietary frontier judge, and only on clean SciFact pairs; the small panel of instruction-tuned models from three developers trails even there, and all open models fall well short on harder, more policy-like claims, where only the proprietary judge approaches human agreement. The openness of the approach is real but confined to the surrogate layer; the high-quality anchor is not yet open, which is the opposite of a "no-proprietary-model" result and we say so. (3) An application that recovers a robust partisan asymmetry in AI engagement and estimates the rate at which each party's claims are *corroborated by the topically-nearest literature*—which is precisely what the instrument measures, and which we distinguish from observed evidence *use*, since legislators do not cite the papers we retrieve. (4) An explicit account of what is and is not validated: the judging step is validated on benchmark pairs in which claim and evidence are already topically matched, but claim extraction and reference retrieval are not, the headline corroboration rates depend on a proprietary generative retrieval step that reproducible open keyword retrieval does not reproduce, and an in-domain human audit—foreclosed by the no-human-coder design—remains the decisive missing test. The result is not a finished autonomous instrument but a validated component, a reproducible open scaffold around it, and an honest map of the distance still to cover.

2 Results

2.1 The pipeline, and where it is open

We instantiate the pipeline of §4 end to end with no human coders. Claim extraction turns each legislative document into a set of empirical claims; for each claim we retrieve candidate references (§4.1 describes the two retrieval configurations); and each (claim, reference) pair is assigned a stance—SUPPORT, REFUTE, or SILENT—by a high-capacity LLM judge, a panel of three open-source models, and two natural-language-inference (NLI) classifiers. The open components—the NLI surrogate, the panel, OpenAlex retrieval, the estimator—run locally on a single consumer GPU with no paid service; the high-capacity judge is a *proprietary* frontier model and is not one of them (§4.2). The substantive question—what fraction of legislators' empirical claims the topically-nearest literature *corroborates*—reduces to aggregating these stance labels with valid uncertainty, which in turn requires that the judge be trustworthy. We establish how trustworthy, and on what, next.

2.2 A proprietary judge approaches the human ceiling; open models lag on the hard task

We score every validator against the expert-human gold on two scientific-claim-verification benchmarks that bracket the policy setting: SciFact (Wadden et al., 2020), expert-written claims paired with their cited abstracts, and Climate-FEVER (Diggelmann et al., 2020), contested real-world claims in lay language paired with retrieved evidence. Both use the identical three-way SUPPORT/REFUTE/SILENT judgment our pipeline makes. To keep the comparison like-for-like, Table 1 reports accuracy with Wilson 95% intervals on the *same* pairs the proprietary judge labelled ($n = 66$ and 45); the judge sample is small by design, mirroring its gold-on-a-sample role, so the intervals are wide and we read them accordingly (Fig. 1).

On clean SciFact pairs the proprietary judge reaches 0.80 accuracy $[0.69, 0.88]$, but the open NLI surrogate is *not separated* from it at this sample size (0.76 and 0.74 vs. 0.80, intervals overlapping)—with $n = 66$ the comparison is underpowered to resolve even a sizeable gap, so this is non-resolution,

Table 1: Accuracy (Wilson 95% CI) of every validator on the *same* pairs the proprietary judge labelled—$n = 66$ on SciFact, $n = 45$ on Climate-FEVER—so the comparison is like-for-like rather than across different samples. On clean SciFact pairs the open NLI surrogate is statistically not separated from the proprietary judge at n=66 (overlapping intervals mean the gap is unresolved, not that the methods are equivalent); the judge's advantage appears only on the harder, more policy-like Climate-FEVER claims. Bottom rows compare the judge's Cohen's κ with the gold to the human inter-annotator agreement each benchmark reports.

Validator (open unless noted)	SciFact (n=66)	Climate-FEVER (n=45)
Claude Opus 4.8 (*proprietary*)	0.80 [0.69, 0.88]	**0.78 [0.64, 0.87]**
NLI DeBERTa-MNLI	0.76 [0.64, 0.84]	0.49 [0.35, 0.63]
NLI BART-MNLI	0.74 [0.63, 0.83]	0.51 [0.37, 0.65]
Qwen2.5-3B-Instruct	0.70 [0.58, 0.79]	0.56 [0.41, 0.69]
Phi-3.5-mini-instruct	0.53 [0.41, 0.65]	0.67 [0.52, 0.79]
OLMo-2-7B-Instruct	0.53 [0.41, 0.65]	0.51 [0.37, 0.65]
TF–IDF (lexical floor)	0.44 [0.33, 0.56]	0.42 [0.29, 0.57]
Judge κ vs. gold	0.71	0.67
Human inter-annotator agreement	$\kappa = 0.75$	$\alpha = 0.33$

not demonstrated equivalence; the small open-panel models trail (0.53–0.70) and the TF–IDF floor is 0.44. In Cohen's κ against the gold the judge scores 0.71—just below the $\kappa = 0.75$ agreement SciFact reports among its expert human coders (Wadden et al., 2020), so "approaching human agreement," not "human-level," is the accurate phrase. The judge's REFUTE detection is strong (precision 0.85–1.0 across benchmarks), answering the worry that the pipeline cannot recognise disconfirming evidence—though, as §2.4 notes, retrieval rarely surfaces such evidence in the application.

The judge's advantage appears on Climate-FEVER, the benchmark closest to legislative communication. There it holds at 0.78 accuracy $[0.64, 0.87]$ while the NLI classifiers fall to 0.49–0.51, barely above the 0.34 TF–IDF/chance floor for this three-class task and far below the judge (intervals do not overlap). Strict entailment between a lay claim and a noisy evidence sentence is uninformative—exactly the regime the policy corpus occupies. Tellingly, human annotators themselves agree only weakly on Climate-FEVER (Krippendorff's $\alpha = 0.33$; Diggelmann et al., 2020), and the judge agrees with the aggregated gold at $\kappa = 0.67$, at least as consistently as the humans agree with one another; it also handles the off-topic NOT-ENOUGH-INFO class well (recall 0.87), which is the loosely-related-pairing regime the application produces. This is why we anchor estimation on the judge and treat NLI as a surrogate to be corrected (§4.4)—and equally why the open-only configuration is not yet a substitute for the judge on hard, policy-like claims. We flag, however, that this central distinction rests on a small judge sample ($n = 45$, 15 per class, with cells as thin as a support precision of 1.0 from 11/11); enlarging the Climate-FEVER judge set is the cheapest way to firm up the validation claim that carries the most weight, and we mark it as the first robustness step.

2.3 Self-preference does not explain the judge's accuracy

The benchmark results rule out the central circularity threat. SciFact and Climate-FEVER claims and evidence are written by humans, so a judge that labels them correctly is assessing the science, not recognizing model-generated prose: self-preference (Panickssery et al., 2024) cannot manu-

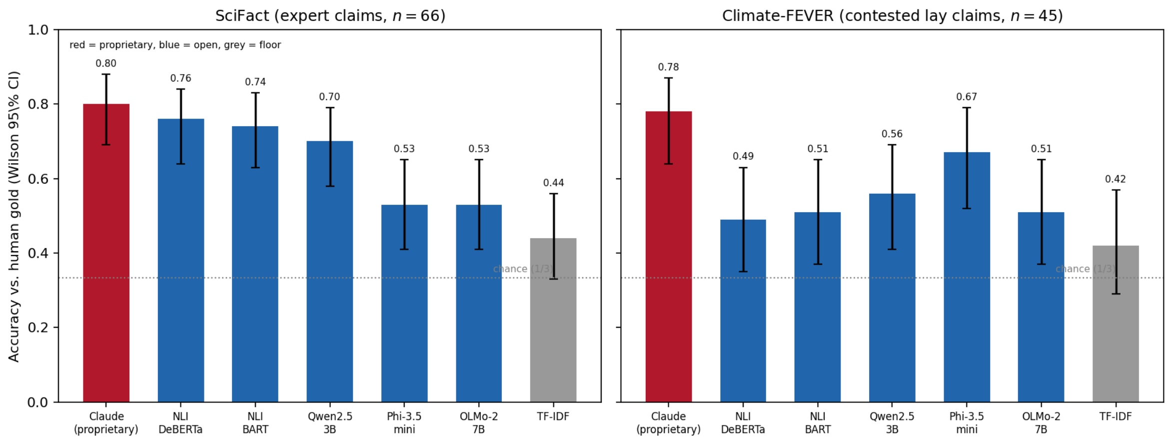

Figure 1: Accuracy with Wilson 95% intervals on the *same* pairs the proprietary judge labelled (red), the open validators (blue), and the lexical floor (grey). On clean SciFact pairs the judge and the open NLI surrogate have overlapping intervals; on the harder, policy-adjacent Climate-FEVER claims the judge's interval clears the open models'. Both tasks are three-class (chance = 1/3).

facture an 80% match to human labels on human text (Fig. 2), and it cannot explain the judge's accuracy on the policy-adjacent Climate-FEVER claims either. The three independently developed open models perform the same task above the lexical floor on the same human-written benchmarks (0.53–0.70 on SciFact, 0.51–0.67 on Climate-FEVER; Table 1), confirming that the task is tractable for models that did not generate the text. We do *not*, however, lean on the panel as an independent guarantee against self-preference: independently *developed* is not independently *erring*—models trained on overlapping corpora and aligned by similar methods share systematic tendencies, and the panel's agreement with the gold is itself only modest ($\kappa = 0.28$–0.44 against the gold, across both benchmarks). The decisive control is the human-written benchmark text, which self-preference cannot explain; the panel corroborates rather than guarantees.

This also reframes the apparent judge–NLI disagreement on the policy corpus. There the judge and NLI agree only weakly ($\kappa = 0.15$–0.21), which one might read as a problem for the judge. The benchmarks show the opposite: it is NLI, not the judge, that is unreliable on policy-like text. Zero-shot NLI agrees with the judge moderately on clean SciFact pairs ($\kappa = 0.46$–0.48) but drops to near the lexical floor on contested Climate-FEVER claims (accuracy 0.49–0.51 against a 0.34 floor; $\kappa = 0.23$–0.27 with the human gold). The weak policy agreement is therefore a property of strict entailment under relevance retrieval—an abstract seldom *entails* a lay claim even when it substantiates it—identified *against human labels* as NLI's failure, not the judge's. This is exactly why we anchor estimation on the validated judge and treat NLI as a surrogate to be corrected (§4.4), not as a free-standing measure.

2.4 Application: how often the nearest literature corroborates legislators' AI claims

We apply the pipeline to California State Assembly member press releases (2015–2025). What it measures is specific and we name it precisely: legislators do not cite papers, so the instrument *supplies* the evidence, retrieving the topically-nearest literature and asking whether it corroborates each extracted claim. It therefore measures claim–literature *corroboration*, not observed evidence *use*.

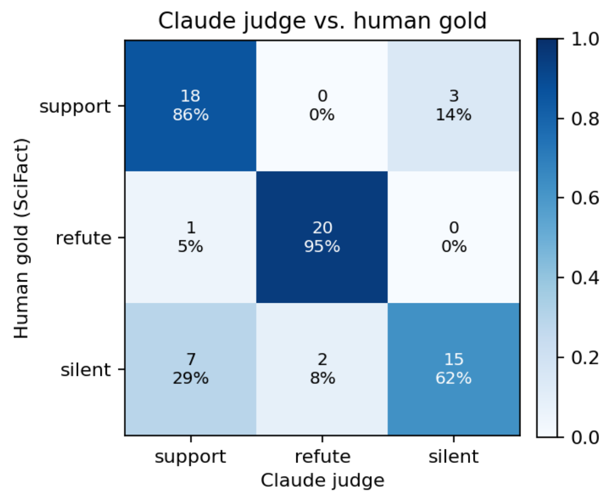

Figure 2: Confusion matrix of the LLM judge against SciFact human gold. Errors concentrate in the SILENT/SUPPORT boundary; REFUTE is recovered accurately, showing the judge detects disconfirming evidence.

Engagement is large and partisan. Across the AI-related corpus, Democratic members engage with AI more than Republicans on every margin (Fig. 3): they mention AI in 4.8% of releases versus 2.6%, and when they do they say more—3.3 AI statements per document on average versus 1.6. These are descriptive aggregates over an overwhelmingly Democratic chamber in which a few prolific offices can drive intensity; we report them without member-clustered intervals and so read the gap as descriptive, not as a tested effect (§3).

Corroboration rates are similar across parties, but this is the least-validated number in the paper. Anchoring on the benchmark-validated judge and correcting the NLI surrogate by prediction-powered inference, the estimated corroboration rate is 0.71 (95% CI $[0.57, 0.84]$) for Democratic claims and 0.54 ($[0.41, 0.67]$) for Republican claims (Fig. 4). The naive NLI estimate is just 0.21 (Democratic) and 0.11 (Republican), so PPI's 0.71 is a roughly threefold rectification driven almost entirely by the 60 judge anchor labels per party: the surrogate is near-constant, so PPI is in effect reporting the judge's mean on a small gold sample (§4.4). We stress where this leaves the headline. It is *conditional on the judge* and not validated against in-domain human labels; worse, it is the one quantity in which all three model stages stack—LLM-*extracted* claims, paired with references a proprietary model *generated* as claim-targeted, scored by an LLM *judge*. That is precisely the all-LLM circularity the paper warns about, and benchmark validation does not reach it: the benchmarks pair human-written claims with human-cited or relevance-retrieved human evidence—the opposite end of the pipeline from where the headline is computed. The corroboration rate is thus the least-validated number we report, not because the judge is weak but because it is where the validated anchor does not reach. The *level* is moreover strongly judge-dependent—across the five validators the Democratic rate ranges from 0.21 (NLI) to 0.79 (a panel model)—which is exactly why a validated anchor matters: it identifies which level to take seriously rather than leaving the analyst to choose. What is robust to that choice is the *ordering*: every validator rates Democratic claims at least as corroborated as Republican, so the modest partisan gap is not an artifact of one judge. But the partisan intervals overlap; the robust asymmetry is in the *volume* of

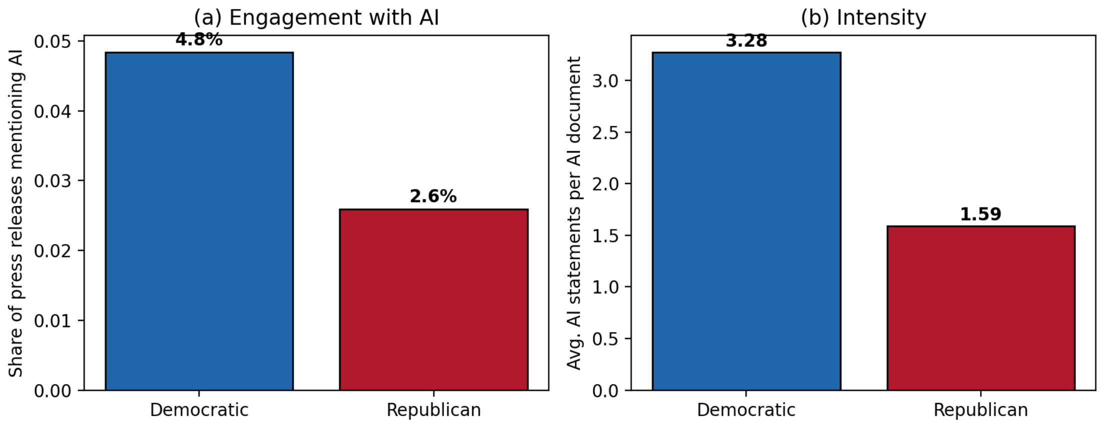

Figure 3: Democrats engage with AI more often (a) and more intensively (b) than Republicans in California Assembly press releases.

AI engagement, not in how well each party's claims are corroborated. We read the level cautiously on two further grounds: relevance-seeking retrieval bounds corroboration from above—it does not search for disconfirming evidence—and press releases are advocacy, so a high rate reflects selective citation of favourable science as much as evidence quality.

2.5 Reproducibility, the retrieval bottleneck, and domain transfer

Reproducible retrieval, and the binding role of retrieval relevance. Replacing the generative reference retrieval of earlier pipelines with retrieval from the open OpenAlex index returns only real, DOI-bearing papers, eliminating the citation-fabrication failure mode of asking a model to name sources. That guarantee, however, applies to OpenAlex and *not* to the generative path the headline rates actually use: only 27–35% of the GPT-5 references carry a DOI, so the references against which the corroboration judgments are made are themselves only partly verifiable—partial exposure to the very failure mode the OpenAlex choice closes. The reproducible path, in turn, exposes the binding constraint on this kind of measurement. OpenAlex keyword search returns broad, topically-related papers rather than the specific study a claim invokes, so across *every* judge—both NLI classifiers and all three open panel models—nearly all pairs are labelled SILENT and corroboration rates fall to a few percent (0.6–13%). The measured *level* is thus governed at least as much by how precisely retrieval targets the claim as by the judge: the 0.5–0.7 rates above reflect bespoke, claim-targeted (generative) retrieval, whereas broad keyword retrieval surfaces little claim-specific evidence at all. This localizes the remaining work precisely. The judge's *labels* are committed and frozen—which makes the downstream analysis reproducible, exactly as frozen human labels would, even though the proprietary judging *process* is not—so it is the *retrieval* step, a fine-tuned open retriever in place of keyword search or generative listing, that must be made both reproducible and claim-targeted. The open pipeline is released to support exactly that. *Domain transfer.* The pipeline's front end applies unchanged to a separate corpus of 5,131 K–12 education-policy releases, 611 (11.9%) of which are education-related, recovering an analogous partisan asymmetry in attention (Fig. 5): Democrats issued 80 releases on social-emotional learning and student mental health versus 1 for Republicans, and 38 on school safety versus 4. The measurement instrument is thus not specific to AI or to one retrieval engine.

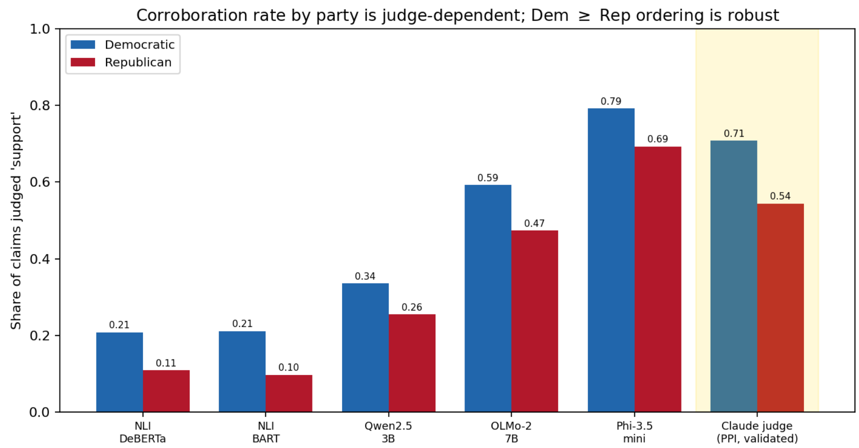

Figure 4: Claim-corroboration rate by party under each judge (generative retrieval). The *level* varies widely across judges (Democratic range 0.21–0.79), but every judge—from zero-shot NLI through each open-panel model to the benchmark-validated judge (gold, PPI-corrected)—rates Democratic claims at least as corroborated as Republican. Estimates are conditional on the judge, not validated against in-domain human labels.

3 Discussion

We set out to measure, at scale, whether the scientific record corroborates the empirical claims legislators make, and to do so in a way that can be *trusted* without new in-domain human labels. The result is partial, and we state it as such. The hardest *judging* step can be validated by transfer from established benchmarks: the high-capacity judge agrees with the expert gold at a level approaching the human inter-annotator ceiling on SciFact ($\kappa = 0.71$ vs the benchmark's 0.75) and, more tellingly, holds up on contested Climate-FEVER claims where the cheap surrogate drops to the lexical floor. Because the benchmark text is human-written, this is not self-preference. But two facts keep the *pipeline* short of a finished autonomous instrument, and they are the spine of the honest account below: the only judge that wins decisively on the hard task is *proprietary*, and the validation exercises the judge on curated pairs rather than the noisier pairings it meets in the application.

Self-preference: bounded on the evidence side, open on the claim side. The most serious objection to an all-LLM loop is that a model judging model-surfaced evidence may exhibit self-preference rather than assess science (Panickssery et al., 2024). Benchmark validation defuses the *evidence*-side channel directly: SciFact and Climate-FEVER evidence is human-written, so a judge that labels it correctly is reading the science, not recognizing its own prose. The cross-developer panel *corroborates* this—small open models from three developers also do the task above chance—but we do not lean on it as an independent guarantee, because models trained on overlapping corpora and aligned by similar methods do not err independently, and their agreement with the gold is itself only modest ($\kappa = 0.28$–0.44). One channel remains genuinely open: in the application the *claims* are LLM-extracted text, so the judge could favour model-generated phrasing on the claim side even when the evidence is human-written, and the human-claim benchmarks do not test this. Self-preference is therefore bounded on the evidence side but not closed on the claim side, and we say so rather than implying the benchmarks settle it.

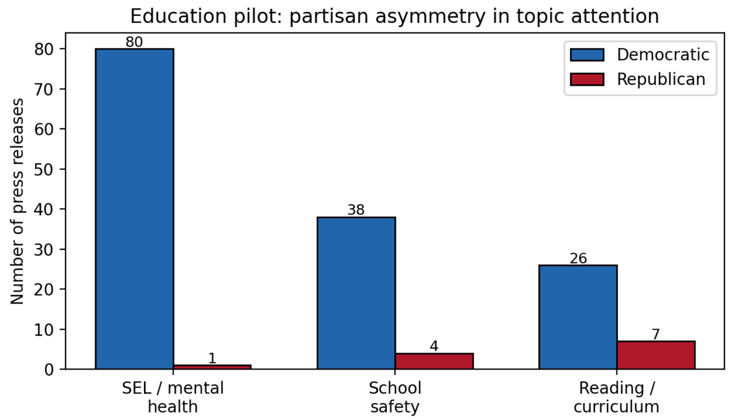

Figure 5: Partisan asymmetry in education-policy attention (counts of press releases), illustrating that the pipeline's front end generalizes beyond AI.

The gap that matters: curated pairs versus application pairs. Our anchor is task validity on benchmark pairs in which the claim and the evidence are already known to be about each other—SciFact pairs a claim with its cited abstract, Climate-FEVER with relevance-retrieved evidence. In the application the judge instead reads claims against keyword hits that are frequently off-topic. The serious residual assumption is thus not subject matter (scientific versus legislative claims) but *pair structure* (curated-relevant versus retrieved-and-possibly-irrelevant). We have partial reassurance—Climate-FEVER's NOT-ENOUGH-INFO class consists of exactly such loosely-related pairings, and the judge handles it—but partial is the right word. The decisive test is an in-domain human audit: a few hundred application pairs coded by people, against which to measure the judge directly. The autonomous, no-human-coder design we were asked to build *forecloses* that audit, which is itself a cost of going fully autonomous; it is the reason we present the application estimates as conditional on the judge and flag the audit as the first thing a follow-up should add. We do not claim the pipeline-as-applied is validated. We claim the judging *operation* is, on benchmark pairs, and that its transfer to application pairs is the open question.

A reusable methodology, honestly scoped. What generalizes beyond AI policy is the *methodology*: validate the judging step cheaply by benchmark transfer, propagate a small set of expensive proprietary-judge labels through a free surrogate by prediction-powered inference, and ground references in an open index so the evidence base is auditable rather than fabricated. Two honest qualifications travel with it. The open surrogate alone is competitive with the proprietary judge on *clean* pairs (SciFact, overlapping intervals) but not on *hard* ones (Climate-FEVER), so the "you don't need a proprietary model" reading holds only where the task is easy. And open keyword retrieval, though exactly reproducible, is not yet precise enough to reproduce the headline rates. The methodology lowers cost and clarifies credibility; it does not yet deliver a fully open, fully validated instrument.

Diagnose across methods. A practical lesson concerns NLI as a stand-alone measure. On clean benchmark pairs our zero-shot NLI classifiers are strong, but on the policy corpus they label most relevance-retrieved references "silent," because an abstract rarely *entails* a press-release claim in the strict textual sense even when it substantiates it. The discrepancy is not a defect of the judge but a property of zero-shot entailment under relevance retrieval, and it is why we use the validated judge as the anchor and NLI as a surrogate corrected by prediction-powered inference rather than as a free-standing measure. Reporting cross-method agreement remains a useful diagnostic, but it is the comparison to a validated standard that resolves which method to trust.

Limitations. Beyond the proprietary-judge and pair-structure issues above, several limits temper the substantive findings. (1) *Retrieval is the binding constraint.* Stance is assessed only over retrieved references with no deliberately disconfirming search, so corroboration rates are an upper and refute rates a lower bound; more importantly, the measured level is highly sensitive to retrieval precision—broad open-index keyword retrieval surfaces little claim-specific evidence and drives nearly all pairs to SILENT (§2.5)—so a reproducible, claim-targeted retriever is the single most valuable component to add. (2) *Claim extraction is unvalidated and could confound the partisan comparison.* An LLM extracts up to ten claims per document; nothing checks that it extracts the right claims or does so at equal rates across writing styles. In fact extraction yield is *not* balanced—about 1.4 extracted claims per Democratic AI release versus 2.4 per Republican release—so the per-claim partisan comparison is confounded by extraction behaviour before the judge ever runs, and a fidelity audit (not done here) is needed to separate the two. (3) *Refute capability is real but nearly idle in the application.* The judge's strong REFUTE detection on the benchmarks answers the worry that the pipeline cannot find disconfirming evidence, but relevance-seeking retrieval suppresses such evidence by construction; it is the retrieval design, not the judge, that limits refute rates here. (4) *Engagement asymmetries lack member-level uncertainty.* The chamber is lopsidedly Democratic and a few prolific offices can drive per-document intensity; the differences in Figure 3 are point estimates over unweighted aggregates and should be read as descriptive, with member-clustered intervals a needed refinement. (5) *Strategic text and scope*: press releases are advocacy, so a high corroboration rate reflects selective citation as much as evidence quality; and the application is one chamber of one state, with extraction and retrieval unvalidated even though the judging step is, so multi-state deployment inherits those gaps rather than only requiring more corpus.

Ethics. Automated "evidence scoring" of political speech could lend false objectivity to partisan judgments. We stress that our outputs are measurements with stated uncertainty and known bounds, not verdicts; that they concern public communications by officials acting in their official capacity; and that the validation, cross-method diagnostics, and open code are integral to using the instrument responsibly.

4 Methods

4.1 The pipeline

The pipeline maps a corpus of political documents to party-level estimates of claim–literature corroboration in three stages, with no human annotation in the loop. We are explicit about paid components as they arise—a proprietary judge and a proprietary generative-retrieval option—rather than claiming the whole pipeline is free.

Claim extraction. Each document is processed by an instruction-tuned LLM prompted to extract up to ten empirical claims—propositions "for which, if you were writing a scientific paper, you would need a reference"—as structured records (document id, claim number, claim text), isolating verifiable propositions from rhetorical or procedural text (Grimmer, 2013).

Evidence retrieval (two configurations). We use and contrast two retrieval configurations and state plainly which produced which numbers. *(i) Generative retrieval*: a proprietary web-connected model (GPT-5 with web search), inherited from the project's earlier pipeline, returns claim-targeted candidate references. This is not reproducible and depends on a paid service, but its targeting is what yields the headline corroboration rates of §2.4; the references it returned are committed as static data. *(ii) Reproducible retrieval*: a query to *OpenAlex* (Priem et al., 2022), a free open index of over 250 million works, restricted to research articles with abstracts, returning title, abstract, DOI, year, venue, and open-access status. OpenAlex is exactly reproducible and every reference it returns is a real, dereferenceable record—eliminating the citation-fabrication failure mode of asking a model to "name references"—but its keyword relevance search returns broader, less claim-targeted papers than the generative step (§2.5). We report results under both and treat the dependence of the support *level* on retrieval precision as a central limitation, not a footnote.

Support assessment. Each (claim, reference) pair is assigned a stance—SUPPORT, REFUTE, or SILENT (irrelevant or insufficient information)—by the validated judges described next. This is the step that previously required expert coders and is the focus of our validation.

4.2 Validators

We treat stance assignment as an annotation problem and use validators that fail in different ways, so that their agreement is informative (Zheng et al., 2023).

- **LLM judge (proprietary).** A high-capacity instruction-tuned model (Claude Opus 4.8), queried in context without fine-tuning, reads the claim and the reference title and abstract and returns a stance label. We are explicit that this is a *proprietary* frontier model: it does not run locally, it is not free, and we do not present it as an open component (see disclosure). Models of this class match or exceed trained human coders on stance tasks (Gilardi et al., 2023; Heseltine and Clemm von Hohenberg, 2024; Törnberg, 2025); whether the *open* alternatives below can replace it is one of our questions.

- **Open-source LLM panel.** Three instruction-tuned models from three independent developers—Qwen2.5-3B-Instruct (Alibaba), Phi-3.5-mini-instruct (Microsoft), and OLMo-2-7B-Instruct (Allen Institute for AI)—run the identical stance prompt locally. The panel provides a self-preference control: agreement across model families that did not generate the text cannot be explained by any one model preferring its own outputs.

- **NLI classifiers.** Two open-source natural-language-inference models with different architectures and training data—`DeBERTa-v3-base-mnli-fever-anli` and `bart-large-mnli`—score each pair with the reference as premise and the claim as hypothesis, mapping ENTAILMENT→SUPPORT, CONTRADICTION→REFUTE, NEUTRAL→SILENT (Yin et al., 2019; Laurer et al., 2024). NLI is the standard discriminative approach to scientific claim verification (Wadden et al., 2020; Koneru et al., 2024) and supplies a cheap surrogate that scales to every pair.

- **Baselines.** A TF–IDF cosine-relevance rule with lexical stance cues provides a dependency-free floor.

All models run locally on a single consumer GPU (Apple M-series, Metal backend); the heaviest component is a 7B-parameter model in half precision.

4.3 Human-anchored validation

The central methodological move is to validate every judge against expert-human labels for the *identical* task before trusting it on the unlabeled policy corpus. We use two benchmarks. SciFact (Wadden et al., 2020) comprises expert-written claims paired with cited abstracts and human SUPPORT/CONTRADICT/NOINFO labels; we form one (claim, abstract) pair per cited document, yielding 340 labeled pairs. Climate-FEVER (Diggelmann et al., 2020) comprises real-world, often contested climate claims with retrieved evidence sentences, each human-labeled SUPPORTS/REFUTES/NOT-ENOUGH-INFO; we evaluate on a class-balanced sample. Both label schemes map onto our SUPPORT/REFUTE/SILENT classes.

We run the NLI classifiers, the open-source panel, and the TF–IDF baseline on every benchmark pair programmatically. The proprietary judge we use deliberately sparsely (mirroring its gold-on-a-sample role): we draw a class-stratified sample, present each (claim, reference) pair *blind to the gold label*, and record its stance—$n = 66$ pairs on SciFact and $n = 45$ on Climate-FEVER. To keep accuracies *comparable*, we additionally evaluate every other validator on the *exact* pairs the judge saw, and report Wilson 95% intervals throughout, because the judge's sample is small. We report accuracy, macro-F_1, per-class F_1, confusion, and pairwise κ. Two scope points are built into the design. First, "human-level" requires a human baseline, so rather than assert it we compare the judge's agreement with the gold to the human inter-annotator agreement the benchmarks themselves report (Cohen's $\kappa = 0.75$ for SciFact; Krippendorff's $\alpha = 0.334$ for Climate-FEVER). Second, these benchmark pairs are *curated to be about each other*—SciFact pairs a claim with its cited abstract, Climate-FEVER with relevance-retrieved evidence—whereas in the application the judge reads claims against keyword hits that may be off-topic. Benchmark accuracy therefore validates the *judging* operation, not its transfer to the noisier application pairings, a gap we treat as the foremost limitation (§3).

4.4 Valid estimation under imperfect labels

Plugging machine labels directly into a downstream analysis biases estimates and invalidates confidence intervals (Egami et al., 2023). We therefore treat the scalable NLI labels as a *surrogate* available for all N pairs and the LLM-judge labels as *gold* on a sample of $n \ll N$ pairs, and combine them with prediction-powered inference (Angelopoulos et al., 2023). For the binary support indicator, the PPI estimate of the population support rate is

$$\hat{\theta}_{\text{PPI}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N} f(x_i)}_{\text{surrogate mean (all pairs)}} + \underbrace{\frac{1}{n}\sum_{j=1}^{n}\big(y_j - f(x_j)\big)}_{\text{rectifier (gold sample)}}, \tag{1}$$

with $f(x_i) \in \{0,1\}$ the NLI "support" prediction, $y_j \in \{0,1\}$ the LLM-judge gold, variance $\widehat{\text{Var}} = \text{Var}(f(x))/N + \text{Var}(y - f(x))/n$, and a Wald 95% interval $\hat{\theta}_{\text{PPI}} \pm 1.96\,\widehat{\text{Var}}^{1/2}$. Equation (1) is the prediction-powered special case of the design-based supervised learning estimator (Egami et al., 2023); both reduce to the human-coded estimate as $n \to N$. Two cautions apply to our application and we state them plainly. First, the gold labeler is the LLM judge, so the intervals are valid for the rate the *judge* would assign over all pairs; their validity for a *human*-equivalent rate rests on the in-domain transfer we cannot yet test (§3). Second, the efficiency gain from the surrogate is small here: on the policy corpus the NLI labels are nearly constant (almost all SILENT), so the surrogate variance term nearly vanishes and the estimate reduces to the judge's mean on the gold sample ($n = 60$ pairs per party). PPI thus functions as valid inference for the judge's labels on a small gold sample, not as a large variance reduction from a strong surrogate, and we present the precision accordingly.

4.5 Corpus and reproducibility

The political application uses a corpus of California State Assembly member press releases (2015–2025), partitioned by the issuing member's party, of which the AI-related subset is analysed here; a parallel K–12 education-policy corpus is used to test transfer of the extraction-and-attention front end. Code, the benchmark harness, the committed judge labels and references, and the figure scripts are public at `https://github.com/bdesmarais/ai-policy-science-use-public`. The open components regenerate from committed inputs with no paid service; the proprietary judge labels and the generative-retrieval references are committed as static data but require their respective proprietary models to regenerate (see Data and code availability).

Data and code availability

Code, the benchmark-validation harness, the committed judge labels, the retrieved references, and the figure scripts are publicly available at `https://github.com/bdesmarais/ai-policy-science-use-public`. We are precise about what is reproducible. The *open* components—the NLI surrogate, the open-source panel, OpenAlex retrieval, the prediction-powered estimator, and all figures—regenerate from committed inputs with no paid service. The two *proprietary* components do not: the gold judge labels were produced by Claude Opus 4.8, and the generative-retrieval references underlying the headline corroboration rates (§2.4) were produced by GPT-5 with web search; both are committed as static data but cannot be regenerated without the respective proprietary models. The validation benchmarks are SciFact (Wadden et al., 2020) and Climate-FEVER (Diggelmann et al., 2020).

AI-assistance disclosure

This measurement instrument uses language models as components, and we disclose exactly which. The stance "gold" labels were produced by Claude Opus 4.8 used as an in-context judge. Claude Opus 4.8 is a *proprietary frontier model*: it does not run locally, it is not free, and it is reached through a paid product; we used it without fine-tuning and without the programmatic API, but we do *not* present it as an open or reproducible component. The scalable surrogate (DeBERTa- and BART-MNLI), the cross-developer panel (Qwen2.5, Phi-3.5, OLMo-2), and OpenAlex retrieval are open and run locally on a single consumer GPU. The same high-capacity model assisted in drafting and in executing the analysis. Every cited reference was checked against a primary source; all design decisions and claims are the authors'. Earlier pipeline development was supported by NSF awards 2318460 and 2148215.

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